$V_{id} = V_+ - V_-$

EECS311 Mid-1

Diodes

Constant model

$\begin{cases} V_D \geq V_{on},\ I_D > 0 \text{ (short)} \\ V_D < V_{on},\ I_D = 0 \text{ (open)} \end{cases}$ ($V_{on} = 0.7V$)

(Ideal model if $V_{on} = 0$)

Exponential model

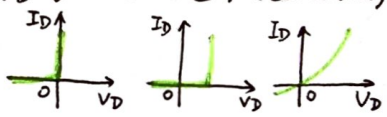
$I_D = I_S(e^{qV_D/nkT} - 1) = I_S e^{V/V_T}$

$q = 1.6\times10^{-19}C$, $k = 1.38\times10^{-23} J/K$

if $n=0$. $V_T = 0.0862T$. (mV)

$= \begin{cases} 25.8mV\ (25^\circ C) \\ 25.3mV\ (20^\circ C) \end{cases}$

$\begin{cases} I_S \text{ doubles } 5^\circ C\uparrow\ (I_R\ 10^\circ C) \\ V_D \downarrow 2mV.\ 1^\circ C\uparrow\ (\text{const } I_S) \end{cases}$



Diode Circuits

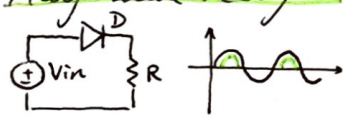
① Assume Region ② Solve

③ Check!!! (也可用图象)

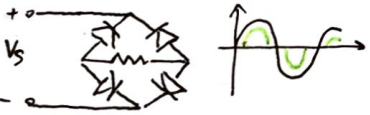
* Constant model (Ideal)

$\begin{cases} \text{不通则看作断路} \\ \text{通则看作反向电压源} \\ \quad (\text{短路}) \end{cases}$

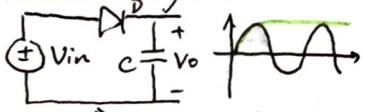
Half-wave rectifier

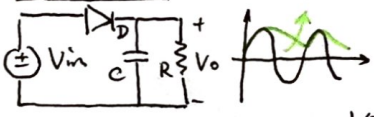


If $V_{in} = k\sin t$, $\langle V_{out}\rangle = \frac{k}{\pi}$.



Peak Rectifier





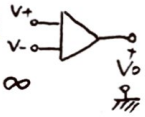
$\Delta t \ll T$, $I_L = V_P/R$, $V_r = \frac{V_P}{fCR}$

$i_{Dav} = I_L(1+\pi\sqrt{2V_P/V_r})$

$i_{Dmax} = I_L(1+2\pi\sqrt{2V_P/V_r})$

$V_o = V_P - \frac{1}{2}V_r$

Ideal Op-amp



① $A_o = V_o/V_{id} \to \infty$

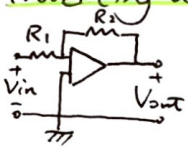
$\Rightarrow V_+ = V_-$

② $R_{in} = \infty \Rightarrow i_+ = i_- = 0$

(output current $\neq 0$)

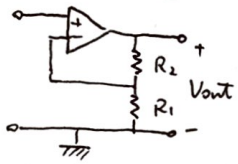
③ $R_{out} = 0 \Rightarrow$ no voltage ↓

Inverting amplifier



$\frac{V_{out}}{V_{in}} = -\frac{R_2}{R_1}$

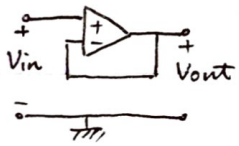
Non-inverting amplifier



$\frac{V_{out}}{V_{in}} = \frac{A_o}{1+A_o f} \xlongequal{A_o\to\infty} \frac{1}{f} = \frac{R_1+R_2}{R_1}$

feedback factor. $:= \frac{V_-}{V_o}$

Voltage follower

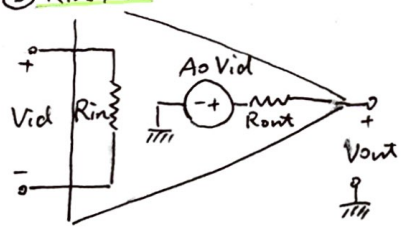


$\frac{V_{out}}{V_{in}} = \frac{A_o}{1+A_o} \xlongequal{A\to\infty} 1$

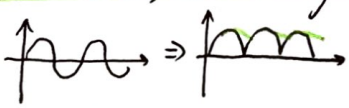
Non-ideal Op-amp

① $A_o \neq \infty$ ② $R_{out} \neq 0$

③ $R_{in} \neq \infty$



(Full-wave) Peak Rectifier



$V_r = \frac{V_P}{2fCR}$.

$i_{Dav} = I_L(1+\pi\sqrt{V_P/2V_r})$

$i_{Dmax} = I_L(1+2\pi\sqrt{V_P/2V_r})$

increased C.

increased R.

decreased T.

A has no impact

④ Common mode rejection ratio (CMRR)

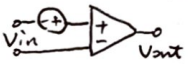
$\begin{cases} V_{id} = V_+ - V_- \\ V_{ic} = (V_+ + V_-)/2 \end{cases}$

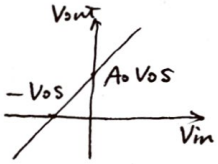
$V_{out} = A_o V_{id} + A_{cm} V_{ic}$

$CMRR = A_o/A_{cm}$

e.g. $\frac{V_{out}}{V_{in}} = \frac{A_o(1+\frac{1}{2}CMRR)}{1+A_o(1-\frac{1}{2}CMRR)}$

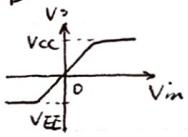
⑤ Input offset voltage



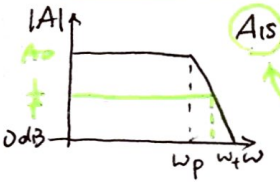


⑥ Input bias current

⑦ Clipping



⑧ Bandwidth



$A(s) = \frac{A_o}{1+s/\omega_p}$

open-loop gain $:= \frac{V_o}{V_{id}}$

If $\omega_t \gg \omega_p$, $\omega_t = A_o\omega_p$

$\Rightarrow f_t = A_o f_p$

unity gain bandwidth = gain-bandwidth product.

⑨ Slew Rate $= \left.\frac{\partial V_{out}}{\partial t}\right|_{max}$

Sallen & Key

$H(s) = \frac{a_2s^2+a_1s+a_0}{b_2s^2+b_1s+b_0}$

$= h\frac{N(s)}{1+d(\frac{s}{\omega_0})+(\frac{s}{\omega_0})^2}$

$d = \frac{b_1}{\sqrt{b_0b_2}}$, $\omega_0 = \sqrt{\frac{b_0}{b_2}}$.

$s_1, s_2 = \omega_0\frac{-d\pm\sqrt{d^2-4}}{2}$

$T_1 = R_1C_1\omega_0$, $T_2 = R_2C_2\omega_0$

$\rho = R_1/R_2$, $\gamma = C_2/C_1$

Bode Plot

① Find $H(s)$ for $s\to\infty, 0$

② Find poles & zeros

	$\lvert H(j\omega)\rvert$	$\angle H(j\omega)$
pole	+20dB/dec	+45°/dec
zero	−20dB/dec	−45°/dec

turning (±idea)

* quadratic case

$\begin{cases} d=0 & \text{undamped} \\ 0<d<2 & \text{underdamped} \\ d=2 & \text{critically damped} \\ d>2 & \text{overdamped.} \end{cases}$

$Q = 1/d$.

Laplace Transform

$F(s) = \mathcal{L}[f(t)] = \int_{t=0^-}^{\infty} f(t)e^{-st}dt$

$f(t) = \mathcal{L}^{-1}[F(s)] = \frac{1}{2\pi j}\int_{\sigma-j\omega}^{\sigma+j\omega} F(s)e^{st}ds$

Initial Value Theorem

$f(0^+) = \lim_{s\to\infty} sF(s)$

Final Value Theorem

$f(\infty) = \lim_{s\to 0} sF(s)$

$u(t) \xleftrightarrow{\mathcal{L}} \frac{1}{s}$

$v(t) = v_0\cos(\omega t+\varphi)$

$= Re\{V_0e^{j(\omega t+\varphi)}\}$

$\Leftrightarrow \hat{V} = V_0\angle\phi$

$V_0 = |V(\omega)| = v_0$

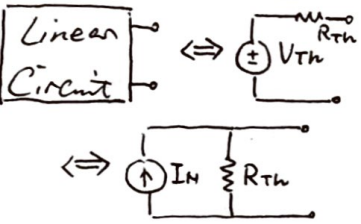
$\phi = \angle V(\omega)$

Euler's Formula:

$\boxed{e^{j\theta} = \cos\theta + j\sin\theta}$

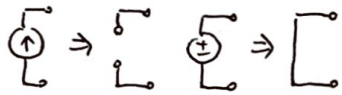
R	$V=IR$	$Z=R$
L	$V=L\frac{di}{dt}$	$Z=sL$
C	$i=C\frac{dv}{dt}$	$Z=\frac{1}{sC}$

Thévenon & Norton



$I_N = V_{Th} / R_{Th}$

* V_{Th}: open-loop voltage
* I_N: short-loop current
* R_{Th}: ① zero out independent sources;



leave dependent ~.

② $R_{Th} = V_T / I_T$

Step Response

For one-stage low-pass filter $V(t) = V_0(1 - e^{-\frac{t}{\tau}})$

rise time: $t_r = \tau \cdot \ln 9$

(time needed from 10% ~ 90%)

NMOS

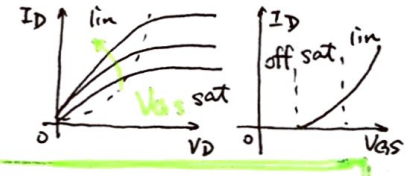
① Cutoff $I_D = 0$. $V_{GS} < V_{Th}$

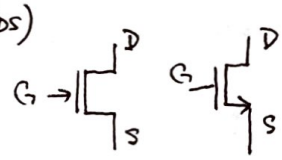
② linear. $V_{GS} > V_{Th}$. $V_{DS} \le V_{GS} - V_{Th} = V_{ov}$

$I_D = K_n [V_{ov} V_{DS} - \frac{1}{2} V_{DS}^2]$

③ saturation $V_{GS} > V_{Th}$. $V_{DS} > V_{ov}$

$I_D = \frac{1}{2} K_n (V_{GS} - V_{Th})^2 (1 + \lambda V_{DS})$





Laplace Transform Properties

$\delta(t) \longleftrightarrow 1$

$\delta(t-T) \longleftrightarrow e^{Ts}$

$1/u(t) \longleftrightarrow \frac{1}{s}$

$u(t-T) \longleftrightarrow \frac{e^{-Ts}}{s}$

$e^{-at}u(t) \longleftrightarrow \frac{1}{s+a}$

$e^{-a(t-T)}u(t-T) \longleftrightarrow \frac{e^{-Ts}}{s+a}$

$t\,u(t) \longleftrightarrow \frac{1}{s^2}$

$(t-T)u(t-T) \longleftrightarrow \frac{e^{-Ts}}{s^2}$

$t^2 u(t) \longleftrightarrow \frac{2}{s^3}$

$te^{-at}u(t) \longleftrightarrow \frac{1}{(s+a)^2}$

$t^2 e^{-at}u(t) \longleftrightarrow \frac{2}{(s+a)^3}$

$t^{n-1}e^{-at}u(t) \longleftrightarrow \frac{(n-1)!}{(s+a)^n}$

$\sin\omega t\, u(t) \longleftrightarrow \frac{\omega}{s^2+\omega^2}$

$\sin(\omega t+\theta)u(t) \longleftrightarrow \frac{s\sin\theta + \omega\cos\theta}{s^2+\omega^2}$

$\cos\omega t\, u(t) \longleftrightarrow \frac{s}{s^2+\omega^2}$

$e^{-at}\sin\omega t\, u(t) \longleftrightarrow \frac{\omega}{(s+a)^2+\omega^2}$

$e^{-at}\cos\omega t\, u(t) \longleftrightarrow \frac{s+a}{(s+a)^2+\omega^2}$

$2e^{-at}\cos(bt-\theta)u(t) \longleftrightarrow \frac{e^{j\theta}}{s+a+jb} + \frac{e^{-j\theta}}{s+a-jb}$

$\frac{2t^{n-1}}{(n-1)!}e^{-at}\cos(bt-\theta)u(t) \longleftrightarrow \frac{e^{j\theta}}{(s+a+jb)^n} + \frac{e^{-j\theta}}{(s+a-jb)^n}$

$\cos(\omega t+\theta)u(t) \longleftrightarrow \frac{s\cos\theta - \omega\sin\theta}{s^2+\omega^2}$

* $\frac{k}{(\beta-\alpha)}(e^{-\alpha t} - e^{-\beta t})u(t) \longleftrightarrow \frac{k}{(s+\alpha)(s+\beta)}$

Laplace Transform

$kf(t) \longleftrightarrow kF(s)$

$k_1 f_1(t) + k_2 f_2(t) \longleftrightarrow k_1 F_1(s) + k_2 F_2(s)$

$f(at), a>0 \longleftrightarrow \frac{1}{a}F(\frac{s}{a})$

$f(t-T)u(t-T) \longleftrightarrow e^{-Ts}F(s),\ T \ge 0$

$e^{-at}f(t) \longleftrightarrow F(s+a)$

$f' = \frac{df}{dt} \longleftrightarrow sF(s) - f(0^-)$

$f'' = \frac{d^2f}{dt^2} \longleftrightarrow s^2F(s) - sf(0^-) - f'(0^-)$

$\int_{0^-}^{t} f(t')dt' \longleftrightarrow \frac{1}{s}F(s)$

$tf(t) \longleftrightarrow -\frac{d}{ds}F(s)$

$\frac{f(t)}{t} \longleftrightarrow \int_s^\infty F(s')ds'$